DATA ANALYSIS CAPSTONE PROJECT REPORT

Spotify User Engagement, Content Performance & Streaming Analytics

Python (EDA & cleaning) · Excel · Power BI dashboard (5 pages) | Prepared by: Vyshnav P S | October 2026

1,000	20,000	5,000	600	49.2%	99,864
Users	Streams	Songs	Artists	Premium	Minutes listened

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How to use this report: every item above and every "Back to Contents" link is clickable (Ctrl + click in Word). Headings also appear in Word's Navigation Pane (View → Navigation Pane).

Project at a glance

Item	Detail
Business goal	Understand listening behaviour, artist popularity, song performance, playlist engagement and streaming trends to improve retention, recommendations, playlist strategy, Premium conversion and content promotion.
Data	Spotify-style workbook with five tables (Users, Artists, Songs, Playlists, Streams): 1,000 users, 600 artists, 5,000 songs, 800 playlists, 20,000 streams.
Method	Cleaning and EDA in Python (pandas, matplotlib, seaborn) → data model and DAX measures in Power BI → five-page interactive dashboard.
Deliverables	Cleaned dataset (.xlsx), cleaning and EDA notebooks (.ipynb), Power BI dashboard (.pbix), this report, presentation.

1. Executive Summary

This project analyses a Spotify-style dataset of **1,000 users, 600 artists, 5,000 songs, 800 playlists and 20,000 streams** to help management improve retention, recommendations, playlist strategy, Premium conversion and content promotion. Data was cleaned and explored in Python, modelled in Power BI and presented through a five-page interactive dashboard.

- **Audience:** users are spread evenly across six countries (Brazil 181, India 178, UK 168, Canada 162, USA 161, Germany 150) with an average age of 38.6. The 35–44 group is the largest (259 users).
- **Engagement:** users listened **99,864 minutes** across 20,000 streams — **4.99 minutes per stream** and about 100 minutes per user.
- **Premium:** 49.2% of users are Premium, but Premium users do **not** listen more than Free users (99.5 vs 100.2 minutes per user, −0.75%, not statistically significant).
- **Content:** Rock (3,959 streams) and Jazz (3,845) together drive 39% of all streams. Artist_339, Artist_277 and Artist_514 are the most consistent hit-makers.
- **Playlists:** followers are spread evenly from 0 to 50K (average 24,918). 157 playlists (20%) are “viral” with 40K+ followers.

Main recommendation: because Premium does not raise listening, conversion should be won with Premium-only features, targeted at the lowest-adoption segments (Canada 43.2%, UK 47.0%, India 48.3%; ages 45–54 at 42.4% and 25–34 at 46.0%).

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2. Business Problem

The company wants to understand **user listening behaviour, artist popularity, song performance, playlist engagement and streaming trends**. Management will use the findings to improve user retention, music recommendations, playlist strategy, Premium subscription conversion and content promotion. The project has five objectives, each answered by specific parts of this report and the dashboard.

#	Objective	Dashboard page	Report section
1	Analyse user listening behaviour	User Analytics · Streaming	5 EDA · 7 Findings
2	Identify top-performing artists and songs	Content Analytics	5 EDA · 7 Findings
3	Evaluate playlist effectiveness	Playlist Analytics	5 EDA · 7 Findings
4	Compare Premium and Free engagement	User · Streaming	7 Findings
5	Recommend ways to increase streaming	Executive Summary	8 Recommendations

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3. Dataset Description

The dataset comes as one Excel workbook with five related tables (raw: Spotify_Dataset.xlsx; cleaned: Spotify_Dataset_Cleaned.xlsx).

Table	Rows	Columns	Description
Users	1,000	User_ID, Age, Country, Subscription	One row per listener; ages 18–59, six countries, Free or Premium
Artists	600	Artist_ID, Artist_Name, Genre	Six genres: Rock, Jazz, Hip-Hop, Classical, Pop, EDM
Songs	5,000	Song_ID, Artist_ID, Popularity, Stream_Count	Stream_Count was added during cleaning (see Section 4)
Playlists	800	Playlist_ID, User_ID, Followers	Owner and follower count; 551 distinct creators

Table	Rows	Columns	Description
Streams	20,000	Stream_ID, User_ID, Song_ID, Minutes_Listened	One row per listening event, 1–9 minutes each

Data model (Power BI). Four many-to-one relationships connect the tables in a chain: Artists ← Songs ← Streams → Users ← Playlists. The Songs–Artists link filters in both directions. The Streams–Songs and Streams–Users links were also set to filter in both directions, so a Country, Subscription or Age slicer reaches songs and artists, and a Genre slicer reaches users.

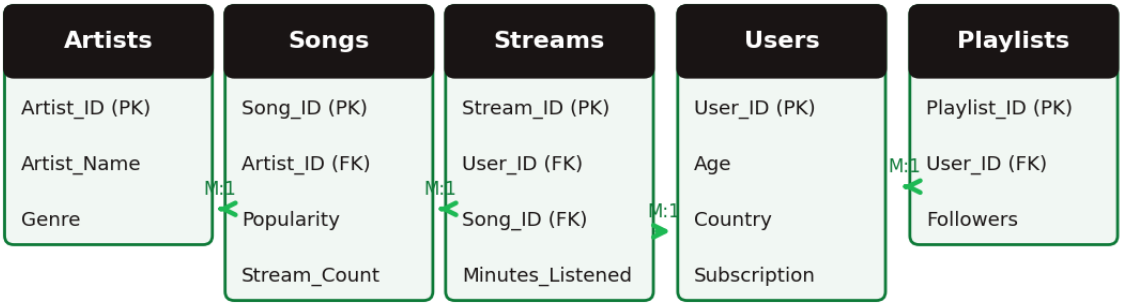


Figure 0 – Data model (arrows point from the many side to the one side)

Relationships: Songs[Artist_ID] → Artists[Artist_ID] · Streams[Song_ID] → Songs[Song_ID] · Streams[User_ID] → Users[User_ID] · Playlists[User_ID] → Users[User_ID].

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4. Data Cleaning

Cleaning was done in Python (pandas) in the notebook 01_data_cleaning.ipynb. The cleaned workbook was then re-checked independently and loaded unchanged into Power BI.

Check	What was done	Result in cleaned data
Duplicate records	Checked ID columns in every table (e.g. Artist_ID)	0 duplicate IDs in all five tables
Missing values	Inspected with .info() on every table	0 missing values
Invalid IDs	Checked that every foreign key matches a parent	0 orphan keys (streams, songs, playlists)
Data types	Reviewed types; text columns identified	IDs as text; Age, Followers, Minutes as integers
Standardise text	Stripped leading/trailing spaces in column names, IDs, names, genres	Six countries, six genres, two subscription values
Stream counts	Counted streams per song, added Stream_Count; songs with none set to 0	Total 20,000 = rows in Streams

Derived field: song popularity

Popularity was **recalculated from streaming activity**: each song's Stream_Count was converted to a percentile rank and scaled to 1–100 (ties share an average rank, results rounded and clipped to 1–100). Popularity therefore measures how a song ranks on streams within this dataset. This has two consequences: popularity and streams move together by design, and "top popular songs" and "consistent hit-makers" describe streaming performance. 79 songs (1.6%) have no streams and sit at the bottom of the scale.

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5. Exploratory Data Analysis

Charts below were produced from the cleaned data; the notebook 02_eda_analysis.ipynb contains the code behind every answer.

Users

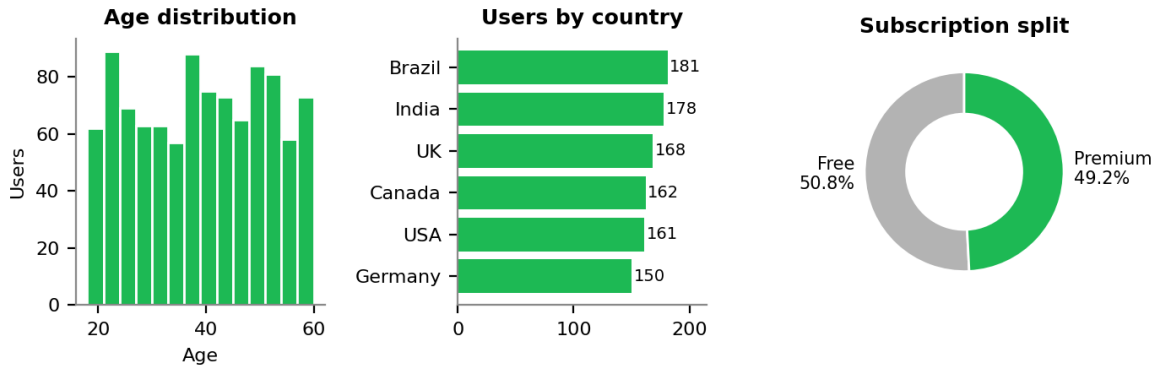


Figure 1 – Age distribution, users by country and subscription split

Artists and songs

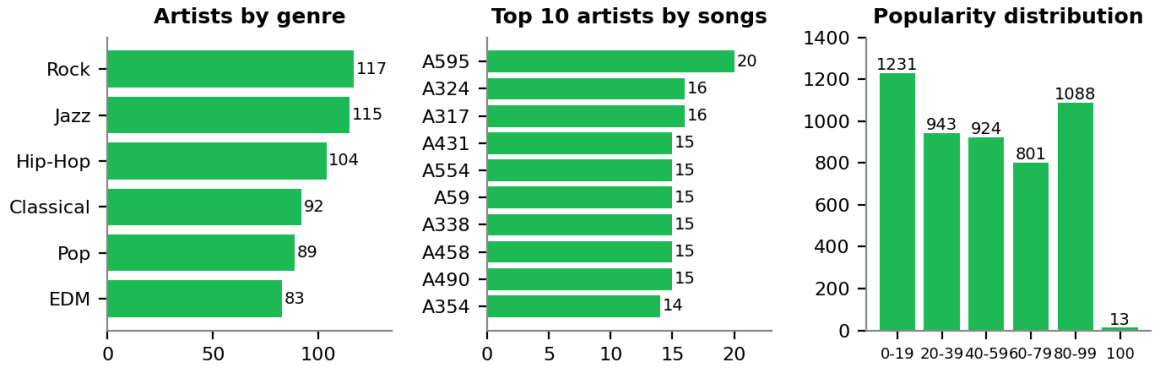


Figure 2 – Genre frequency, artists by song count and popularity distribution

Streaming and playlists

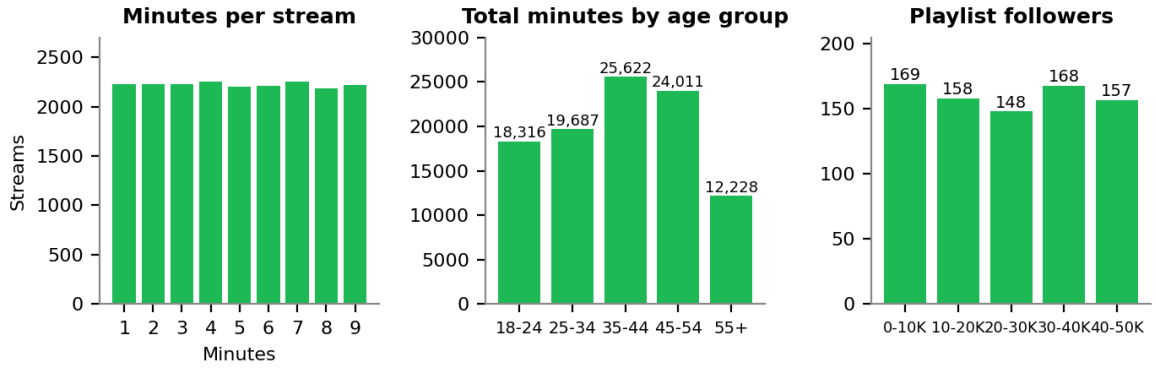


Figure 3 – Listening time per stream, total minutes by age group and playlist followers

Every distribution is close to uniform: listening time per stream is flat from 1 to 9 minutes (about 2,200 streams each), follower counts are spread evenly, and countries and genres differ by only a few percent.

Answers to the 21 business questions

Q	Question	Answer
1	Country with most users	Brazil (181 users)
2	Average age	38.6 years (range 18–59)

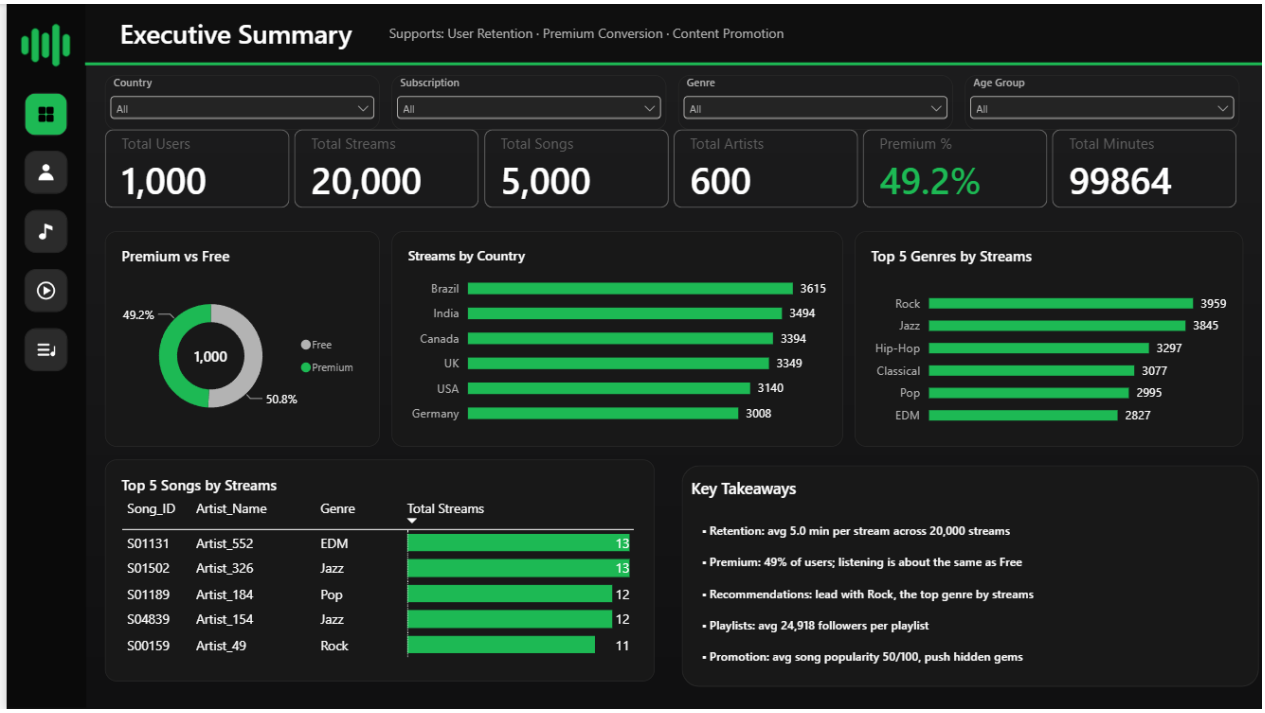
Q	Question	Answer
3	Age group listening most	35–44: 25,622 minutes (largest group; per-user minutes are ≈99–104 in every group)
4	% Premium	49.2% (492 of 1,000)
5	Artists with most songs	Artist_595 (20), then Artist_317 and Artist_324 (16 each)
6	Dominant genres	Rock (977 songs, 117 artists), Jazz (964, 115), Hip-Hop (842, 104)
7	Genres most popular with listeners	Rock 3,959 streams, Jazz 3,845, Hip-Hop 3,297
8	Top popular songs	13 songs tie at popularity 100; S01131 and S01502 lead on streams (13 each)
9	Average popularity	50.10
10	Consistent high-popularity artists	Artist_339 (79.4 avg, 8 songs), Artist_277 (77.3), Artist_514 (75.5); minimum 6 songs
11	Total streams	20,000
12	Most-streamed songs	S01131 and S01502 (13), S01189 and S04839 (12)
13	Users listening longest	U00404 and U00949 (194 min each), U00313 (182)
14	Countries with most listening	Brazil 18,099 min, India 17,518, Canada 16,734
15	Average minutes per stream	4.99
16	Playlists with most followers	P0583 (49,998), P0397 (49,993), P0458 (49,969)
17	What defines successful playlists	Followers only: spread evenly 0–50K (median 25,660); 157 playlists ≥ 40K; no gap by owner subscription (Free 24,986 vs Premium 24,847 avg)
18	Users with most-followed playlists	By total followers: U00464 (172,859 across 4 playlists), U00271 (155,528), U00124 (136,153)
19	Do Premium users listen longer?	No: 99.5 vs 100.2 minutes per user (−0.75%)
20	Highest Premium adoption	Germany 54.0%, USA 52.2%, Brazil 50.8%; lowest Canada 43.2%
21	Engagement difference by subscription	Per stream: Premium 4.96 vs Free 5.03 min (−0.07); total minutes 48,945 vs 50,919

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6. Dashboard Screenshots

The Power BI file (Spotify_dashboard.pbix) has five pages. Each page has slicers for country, subscription, age group and either genre or follower tier, so every view can be filtered interactively.

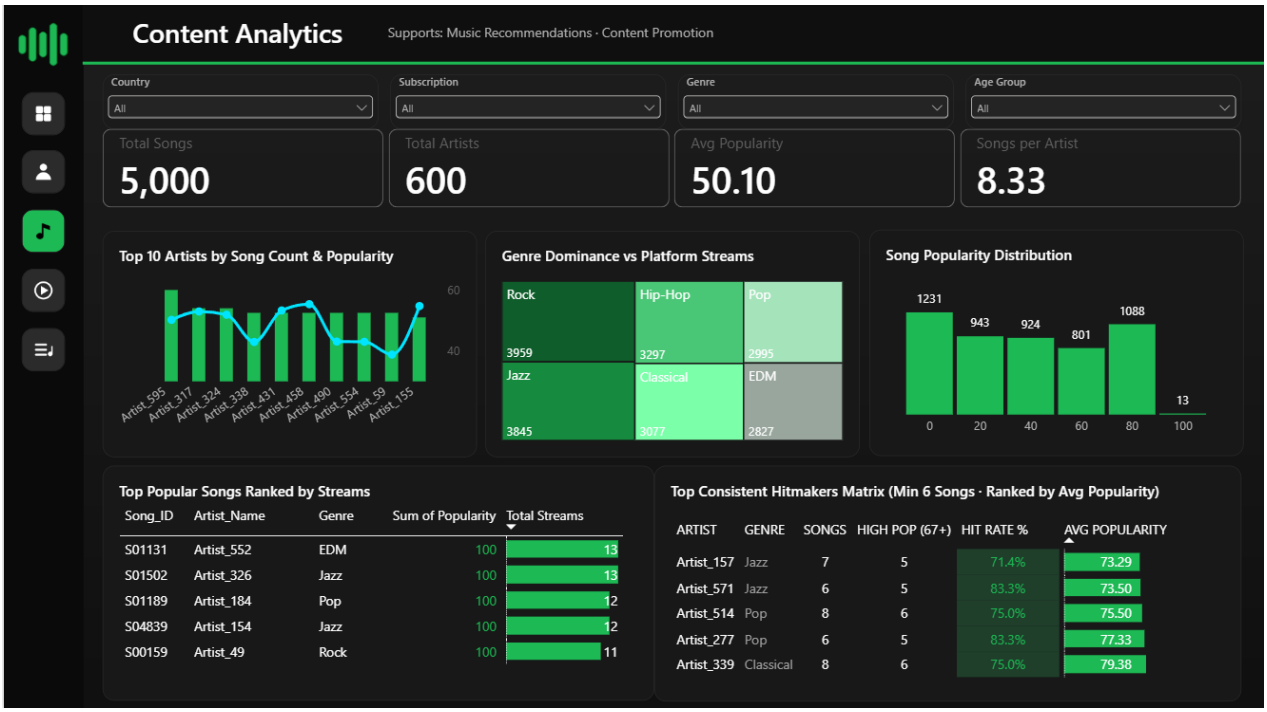
Go to: Executive · Users · Content · Streaming · Playlists



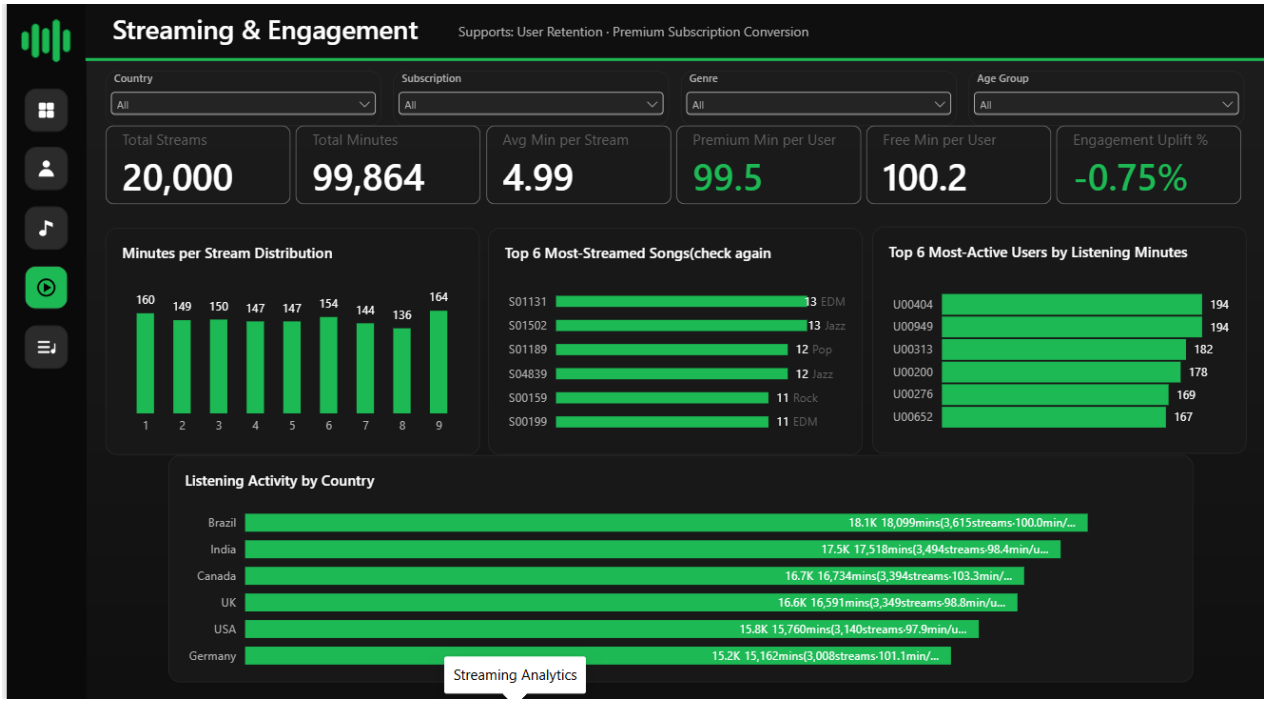
Dashboard 1 – Executive Summary. Six KPI cards, Premium vs Free split, streams by country, top genres, top songs and an auto-generated key-takeaways panel. Supports retention, Premium conversion and content promotion.



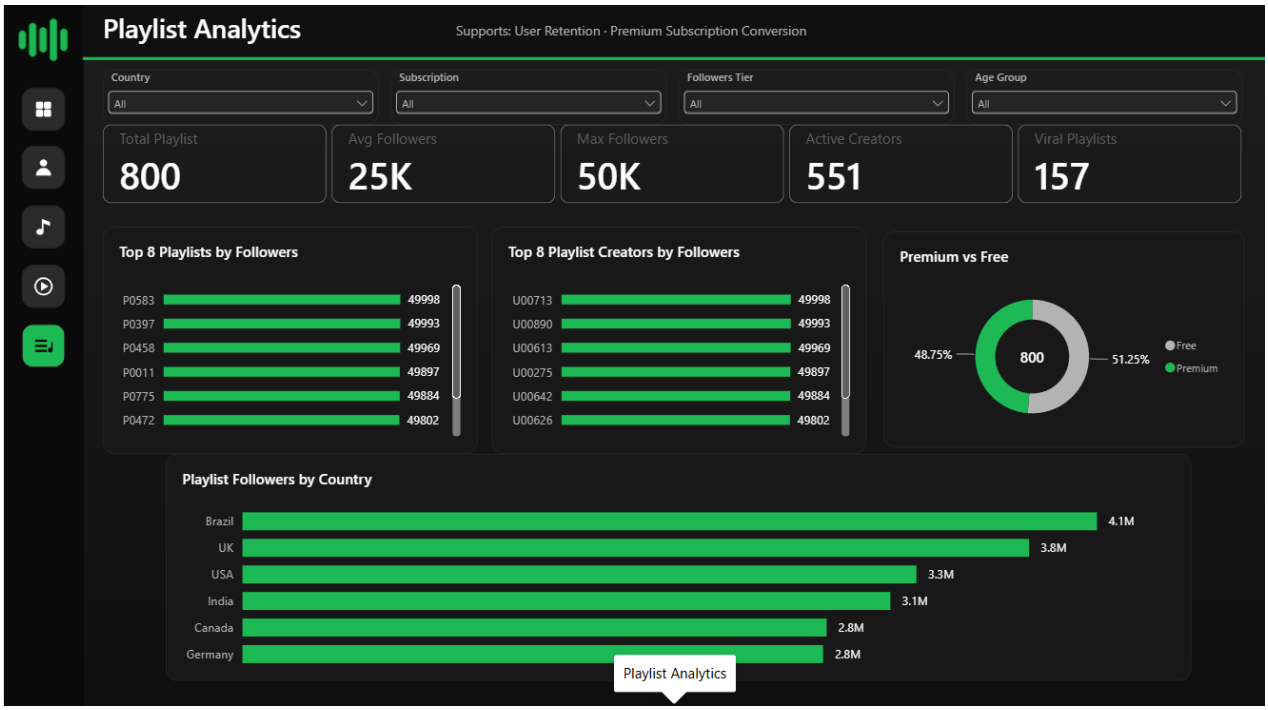
Dashboard 2 – User Analytics. Users by country and age group, Premium share by age and country, minutes per user by age group and subscription. Supports retention and Premium conversion.



Dashboard 3 – Content Analytics. Top artists by song count, genre treemap, popularity distribution, top songs and the consistent hit-makers matrix. Supports recommendations and content promotion.



Dashboard 4 – Streaming & Engagement. Streaming KPIs, minutes-per-stream distribution, most-streamed songs, most active users and listening activity by country.



Dashboard 5 – Playlist Analytics. Playlist KPIs, top playlists and creators by followers, owner subscription split and followers by country.

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7. Key Findings

User behaviour

- **Balanced base.** Six countries each hold 15–18% of users; Brazil leads (181) and Germany is smallest (150). Age is concentrated in 35–54 (502 users, 50%).
- **Listening is flat across segments.** Minutes per user range only from 98.8 (45–54) to 103.6 (55+); by country from 97.9 (USA) to 103.3 (Canada). Total minutes differ mainly because segment sizes differ (35–44 = 25,622 minutes because it has 259 users).

Premium vs Free

- **No engagement uplift.** Premium users average 99.5 minutes versus 100.2 for Free (−0.75%; Welch t-test p = 0.63). Per stream, Premium is 4.96 minutes versus 5.03 (p = 0.07).
- **Adoption differs by segment.** Germany (54.0%) and USA (52.2%) lead; Canada (43.2%) and UK (47.0%) trail. By age, 55+ (55.1%) and 35–44 (54.1%) lead while 45–54 (42.4%) and 25–34 (46.0%) trail.

Content

- **Rock and Jazz lead.** They hold the most artists (117, 115), songs (977, 964) and streams (3,959, 3,845); EDM is smallest on all three (83 artists, 696 songs, 2,827 streams).
- **Top songs are tied.** 13 songs reach popularity 100; S01131 (EDM) and S01502 (Jazz) lead with 13 streams. Seventeen artists qualify as “Consistent Hitmakers” and nineteen as “Hidden Gems” (average popularity ≥ 70 with fewer than 8 songs).
- **Long tail.** 1,231 songs (24.6%) sit in the lowest popularity band (0–19), including 79 with no streams.

Playlists

- **No concentration.** The top three playlists have 49,969–49,998 followers, but followers are spread evenly; 157 of 800 playlists (19.6%) pass 40K.
- **Creators.** 551 of 1,000 users (55%) own a playlist. U00464 has the most total followers (172,859 across 4 playlists). Brazil (4.1M) and UK (3.8M) lead on total followers; Germany (26.7K) and UK (26.4K) lead on average per playlist.

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8. Business Recommendations

Management goal	Recommendation	Evidence	Where
Premium conversion	Target Canada, UK, India and the 25–34 and 45–54 age groups with trial offers. Sell Premium on features (offline, no ads, audio quality) because it does not currently increase listening.	Premium 43.2% Canada, 47.0% UK; 42.4% (45–54); uplift −0.75%	Dashboard 2
User retention	Run re-engagement campaigns in the lowest-engagement segments (USA 97.9, India 98.4, 45–54 at 98.8 min/user). Add date fields so retention can be measured over time.	Engagement flat at ≈100 min/user	Dashboards 2, 4
Music recommendations	Lead recommendation carousels with Rock and Jazz (39% of streams). Use Hip-Hop (16.5%) as the third pillar and test EDM, which has similar popularity but the fewest streams.	Rock 3,959; Jazz 3,845; EDM 2,827	Dashboard 3
Playlist strategy	Feature the 157 viral playlists and partner with top creators (U00464, U00271, U00124). Activate the 449 users who own no playlist with “create your first playlist” prompts.	157 playlists ≥ 40K; 551 active creators	Dashboard 5
Content promotion	Promote the 19 “Hidden Gems” artists and the 17 consistent hit-makers; review the 79 never-streamed songs and the 1,231 songs in the lowest band.	Artist_339 79.4 avg popularity	Dashboard 3

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9. Conclusion

The analysis answers all 21 business questions and covers every KPI in the project guide through a five-page interactive dashboard. The platform has a balanced, evenly engaged audience: no country, age group, genre or subscription tier stands far above the rest. The clearest opportunities are converting low-adoption Premium segments, leading recommendations with Rock and Jazz, and activating non-creators and hidden-gem artists.

Limitations

- **No date column,** so streaming trends over time and true retention cannot be measured.
- **Popularity is derived from streams** (Section 4), so it is not an independent quality score.
- **Near-uniform data** means gaps between segments are small; most should be treated as directional, not proven.
- **Playlists are not linked to songs,** so playlist success can only be judged by follower counts and owner traits.

Next steps

- Add timestamps to enable trend, cohort and churn analysis.
- Add playlist–song links to learn what content makes playlists succeed.
- A/B test the Premium offers for Canada, UK and India.

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